

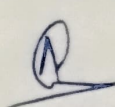


Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY
(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)
Department Of Artificial Intelligence & Machine Learning Engineering

Program Code : AI5K
Name of the Student : Sanika Dhanaji Chougale
Name of the Mentor (Faculty) : Mrs. Poonam Kumbhar
Name of the Mentor (Industry): Mr. Abhijit Gatade
Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week <u>10</u>	Mon, Date <u>04/08/25</u>	Understanding project and dataset features.	Understood project aim and dataset.	 
	Tue, Date <u>05/08/25</u>	Data cleaning and preprocessing step.	Cleaned and prepatterns & corrections	
	Wed, Date <u>06/08/25</u>	Performing exploratory data analysis	Explored data patterns and corrections.	
	Thu, Date <u>07/08/25</u>	Building logistic analysis regression model	Trained and evaluated logistic regression	
	Fri, Date <u>08/08/25</u>	Implementing random forest classifier	Built and turned random forest model	
	Sat, Date <u>9/08/25</u>	Saving final model and reflections.	Saved model and documented workflow.	



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Week No: 10

Day: Monday

Date: 04/08/25

- Understanding Project and Dataset.

The project on crop yield prediction begins with understanding its importance in agriculture and food security. Crop yield refers to the amount of crop produced per unit area and is influenced by multiple environmental and agricultural factors such as rainfall, temperature, soil nutrients, fertilizer usage, and crop type. In this project, the dataset includes both numerical both numerical features (e.g. rainfall, NPK values, temp). and categorical features. The target variable is crop yield, usually measured in kilograms per hectare. Theoretical understanding of the dataset is crucial because machine learning models rely on patterns.



Internships Daily Diary (2025-26)

Week No: 10

Day: Tuesday

Date: 05/08/25

- Data Cleaning and preprocessing

Data cleaning is critical step because real-world agricultural datasets are often incomplete, inconsistent, or noisy. Missing values in rainfall or soil nutrients need to be addressed using imputation techniques such as mean, median, or interpolation. Outliers in fertilizer use or yield values must be identified because they can skew predictions. Categorical variables like crop type or soil type are transformed into numerical form through encoding methods such as one-hot encoding. Numerical features like rainfall and fertilizer are normalized or scaled.



Internships Daily Diary (2025-26)

Week No: 10

Day: Wednesday

Date: 06/08/25

• Exploratory Data Analysis

Exploratory Data Analysis is the process of examining the dataset through visualization and statistical method to uncover hidden patterns.

It allows researchers to understand how different features influence crop yield. Scatterplots can be used to visualize relationships such as rainfall versus yield. Histograms show the distribution of soil nutrient levels, while boxplots highlight variability and outliers in crop production across regions. A correlation matrix identifies the strength of relationships among variables, which is crucial for selecting features that significantly affect yield.



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Week No: 10

Day: Thursday

Date: 07/08/25

• Building Linear Regression Model.

Linear Regression is the most basic Supervised learning algorithm used to predict Continuous Variable such as crop yield. It establishes a linear relationship between independent Variables and the dependent variables.

The mathematical equation takes the form $y = ax + b$, where coefficients represent the influence of each factor on yield.

During training, the model fits the best possible line that minimize the error between actual and predicted values, performance is evaluated using metrics such as the R^2 Score, Mean Absolute Error (MAE), Root Mean Square error.



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Week No: 10

Day: Friday

Date: 08/08/25

- **Implementing Random Forest Model**

Random forest is an advanced ensemble learning technique that overcomes the limitations of Linear Regression by handling nonlinear and complex interactions between variables. It works by constructing multiple decision trees during training and combining their result to improve accuracy and reduce overfitting. Each tree considers different subsets of features, ensuring robustness in prediction. Hyperparameters such as the number of trees, maximum depth, and minimum samples per split are tuned to achieve the best performance.



Internships Daily Diary (2025-26)

Week No: 10	Day: Saturday	Date: 09/08/25
• <u>Saving final Model and Reflections:</u>		
Once the best - performing model is developed , it is saved using tools like joblib or pickle for future use and deployment. Saving the model ensures that it can be integrated into applications for real - time predication without retraining. A complete documenta- tion of the workflow is prepared , covering dataset prepmssing , exploratory analysis, model building , evaluation and result. finally , the completed project lays the foundation for further improvements using neural networks or deployment in web - based platforms.		